Low-E filter

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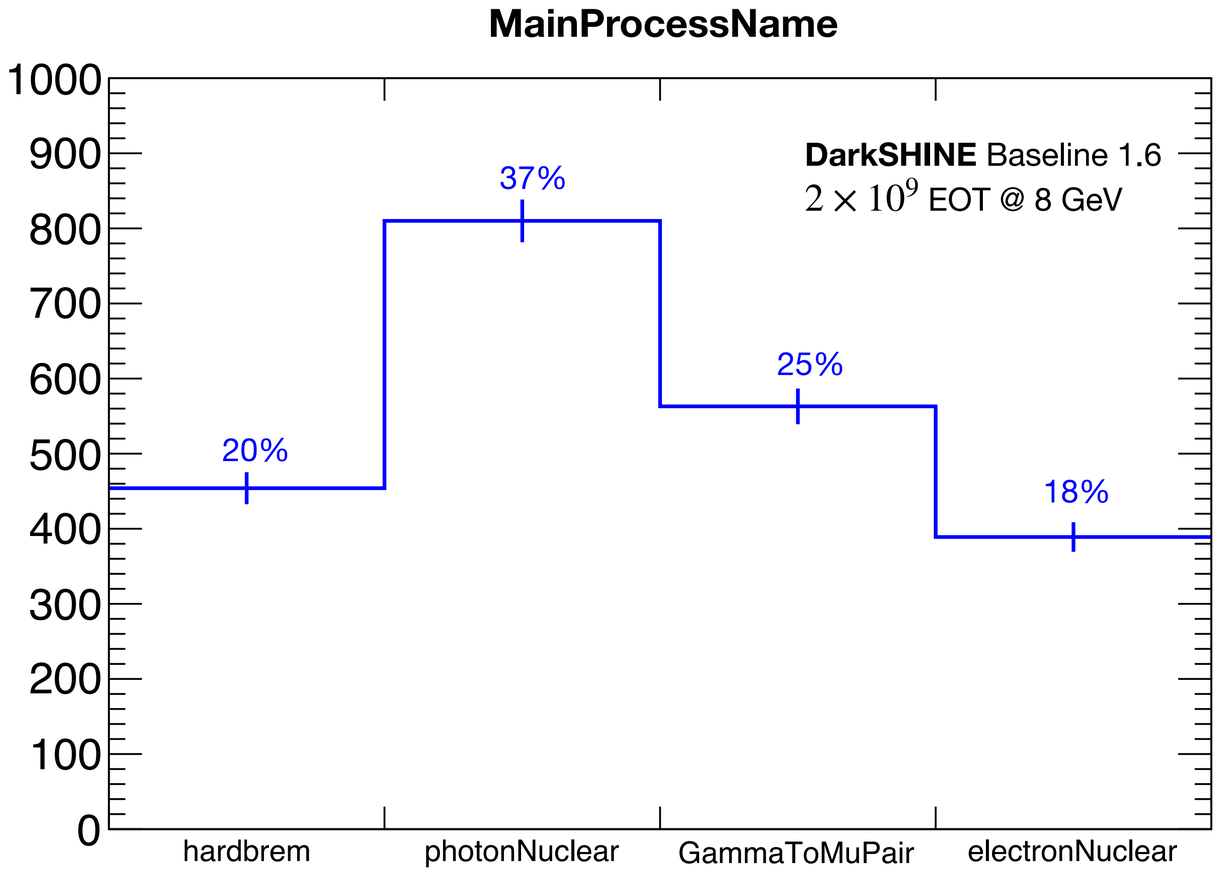
2023/05/17

Low-E filter

description and performance

- This filter will be applied on the DSimu inclusive production. Can also apply to rare process production.
- **This filter consists of 2 filters:**
  - filter1: initial particle (e-) at ECAL surface > 4 * GeV, kill
  - filter2: ECAL deposit energy > 4*GeV, kill
- **Filter performance** (test on M1 MacOS, Baseline 1.6):
  - simulation speed: x20 (event time w/o filter 44ms/EOT → event time with filter: 2.27ms/EOT)
    - 10^{12} EOT ~ 19.7 hours x 32000 threads ~ cost 32000 RMB
  - storage : 1.1×10^{-6} (saved/processed = 2216 / 2e9; event size = 36KB)
    - 10^{12} EOT ~ 38 GB

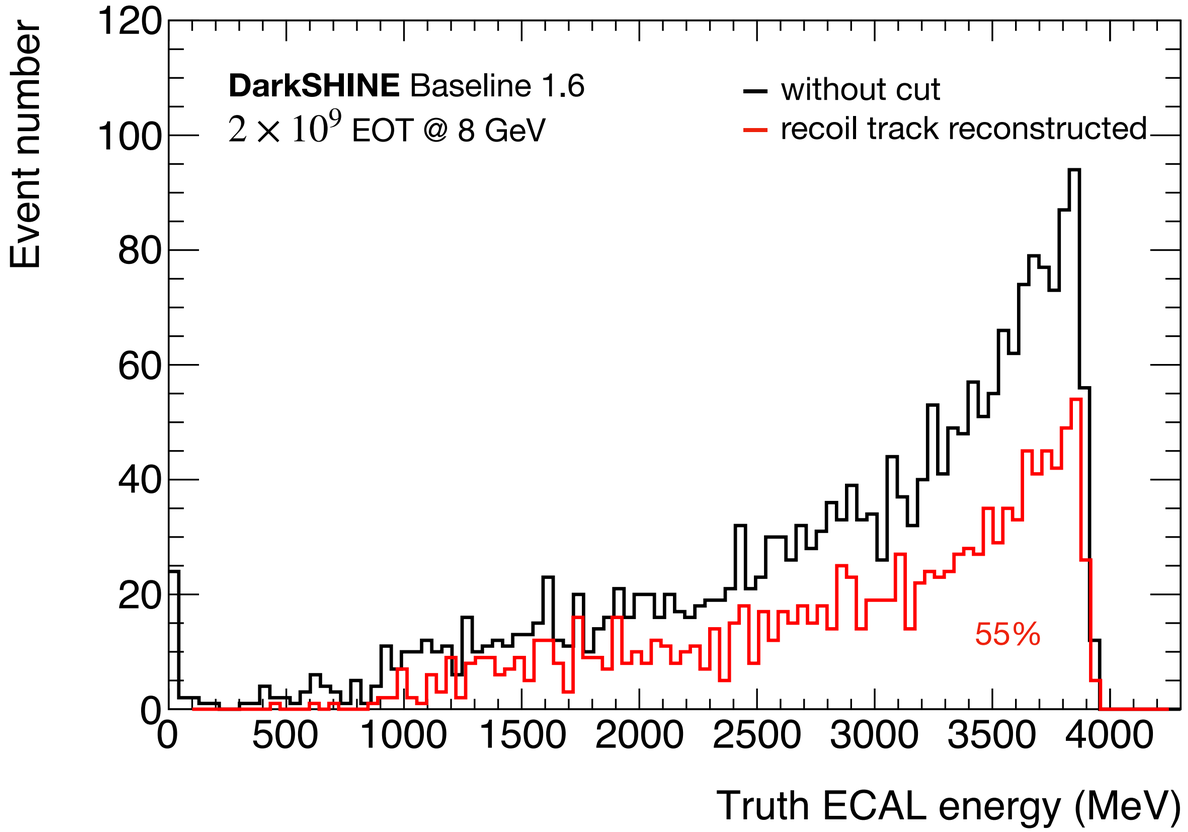
Truth Process

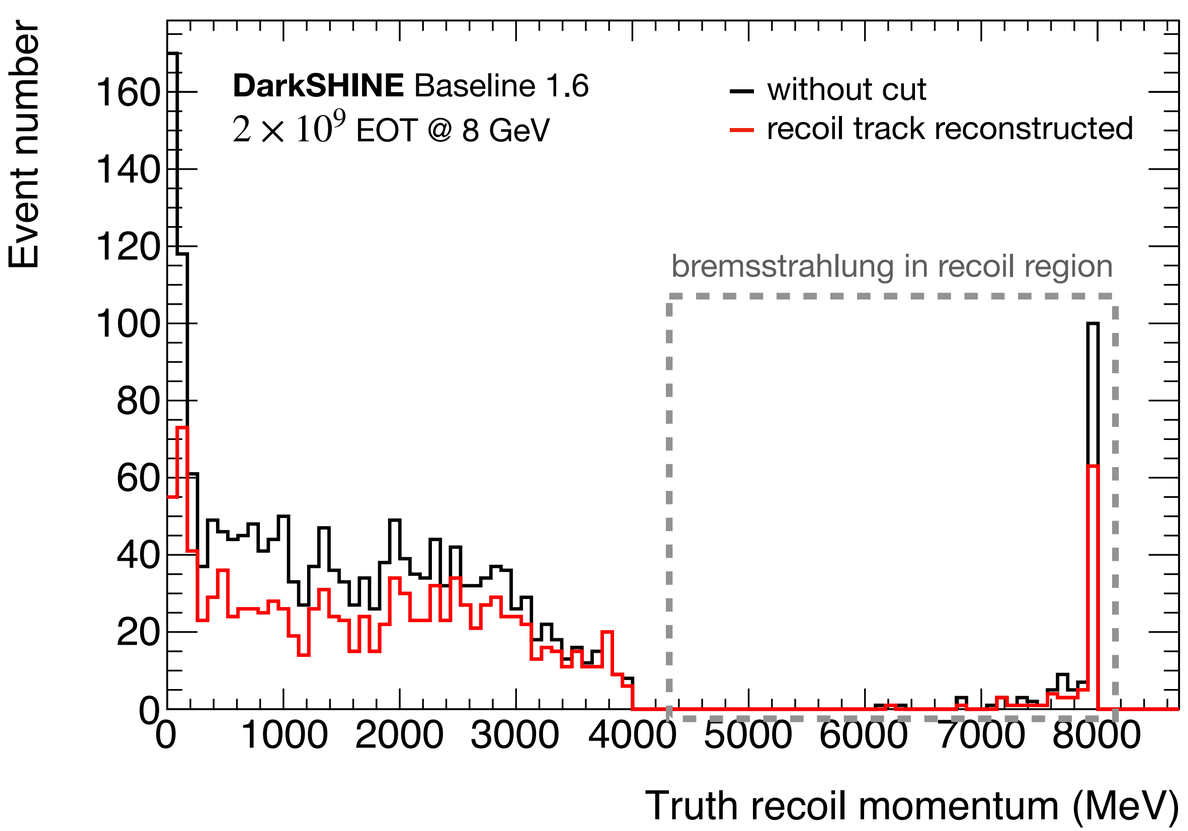


Process Number (not equal to event number)

Process/Volume	Target	ECAL
GMM	21	542
HardBrem (Eγ>4GeV)	1828	66
EN	388	2
PN	116	694

Truth ECAL Energy, Recoil e- Momentum

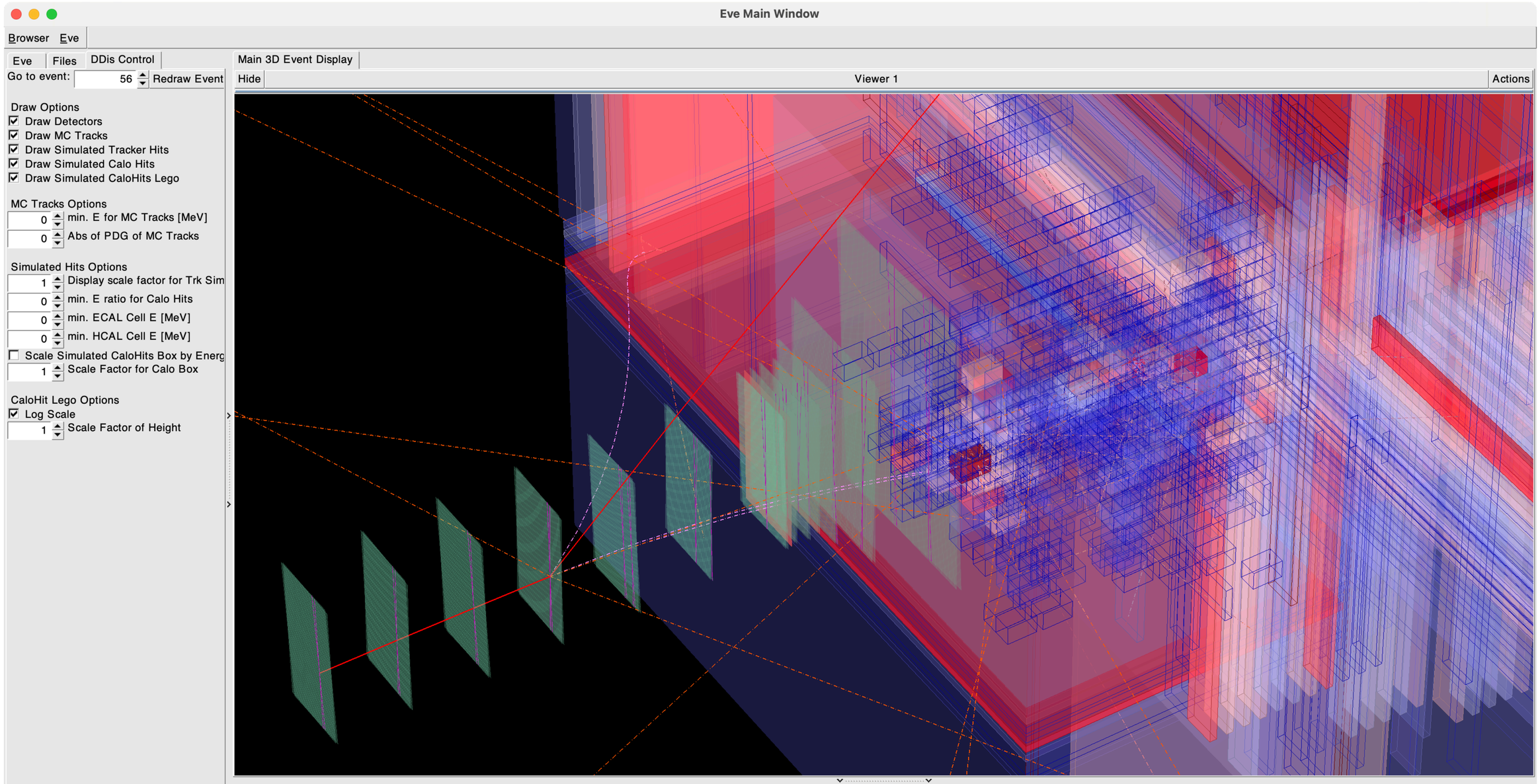




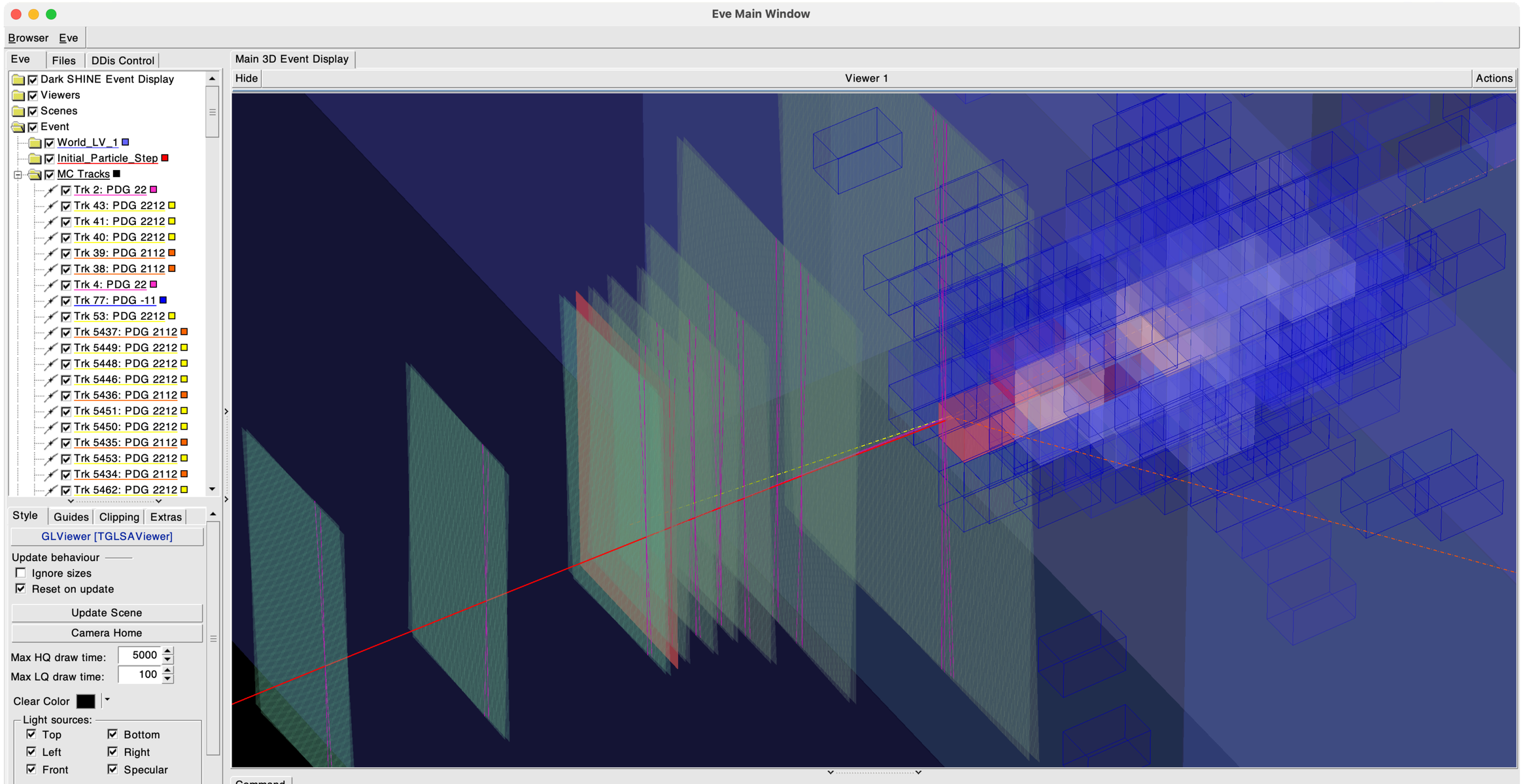
mis-reconstructed track events

	exist truth recoil track	no truth recoil track
reconstructed	54.7%	0.7%
not reconstructed	30.4%	14.2%

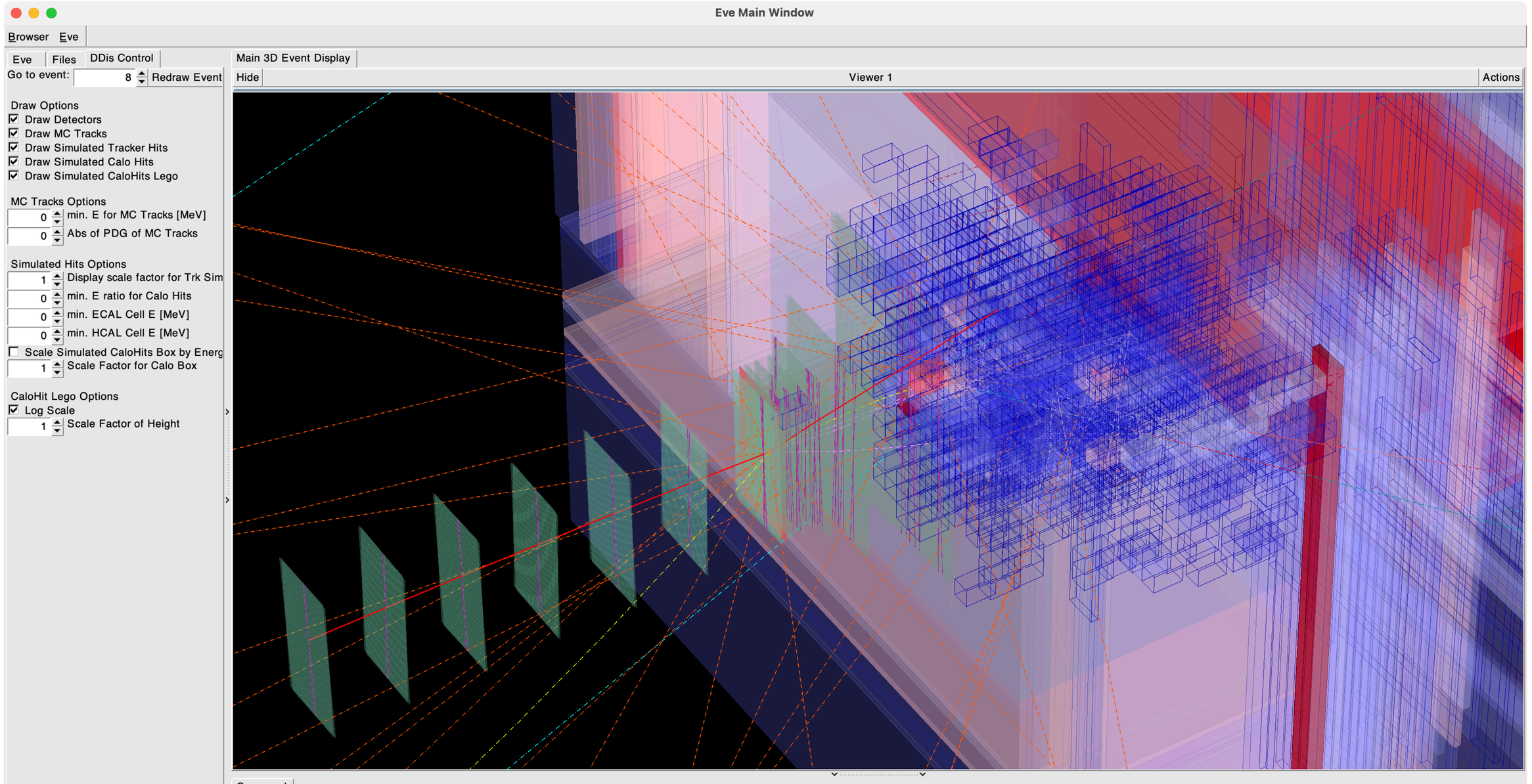
EN/PN, No recoil track, reconstructed (0.7%)



hardbrem/EN/PN/GMM, exists recoil track, not reconstructed (30.4%)



hardbrem/EN/PN/GMM, exists recoil track, not reconstructed (30.4%)



hardbrem/EN/PN/GMM, No recoil track, not reconstructed (14.2%)

